

LI.

What are and what should the numbers be?

Continuation and completion: §§ 8–14 and explanatory note

English translation from the source-checked German, GMW III, pp. 368–391.

§ 8. Finite and infinite parts of the sequence of numbers

119. Proposition. Every system Z_n in 98 is finite.

Proof by complete induction (80). For

$\rho.$ the proposition is true for $n = 1$ by 65 and 102.

$\sigma.$ If Z_n is finite, then by 108 and 70 $Z_{n'}$ is also finite, as was to be proved.

120. Proposition. If m, n are different numbers, then Z_m, Z_n are non-similar systems.

Proof. For symmetry's sake we may by 90 assume that $m < n$; then by 106 Z_m is a proper part of Z_n , and since by 119 Z_n is finite, by 64 Z_m and Z_n cannot be similar, as was to be proved.

121. Proposition. Every part E of the sequence of numbers N that has a greatest number (111) is finite.

The proof follows from 113, 119, and 68.

122. Proposition. Every part U of the sequence of numbers N that has no greatest number is simply infinite (71).

Proof. If u is any number in U , then by 117 there is in U one and only one next greater number than u , which we shall denote by $\psi(u)$ and regard as the image of u . The mapping ψ of the system U that is completely determined in this way evidently has the property

$$\alpha. \quad \psi(U) \subset U,$$

that is, U is mapped by ψ into itself. If, further, u, v are different numbers in U , then for symmetry's sake we may by 90 assume that $u < v$; then from the definition of ψ and by 117 it follows that $\psi(u) \leq v$ and $v < \psi(v)$, hence by 95 $\psi(u) < \psi(v)$. Therefore by 90 the images $\psi(u), \psi(v)$ are different, that is,

$\delta.$ the mapping ψ is similar.

Furthermore, let u_1 denote the smallest number (96) of the system U . Then every number u contained in U is $\geq u_1$, and since generally $u < \psi(u)$, by 95 $u_1 < \psi(u_1)$; hence by 90 u_1 is different from $\psi(u_1)$, that is,

$\gamma.$ the element u_1 of U is not contained in $\psi(U)$.

Thus $\psi(U)$ is a proper part of U , and consequently U is an infinite system by 64. Now, in agreement with 44, if V is any part of U , let $\psi_0(V)$ denote the chain of V corresponding to the mapping ψ ; we must finally still show that

$$\beta. \quad U = \psi_0(u_1)$$

holds. Indeed, since every such chain $\psi_0(V)$ is, by its definition (44), part of the system U mapped by ψ into itself, it is self-evident that $\psi_0(u_1) \subset U$. Conversely, it

follows at once from 45 that the element u_1 contained in U is certainly contained in $\psi_0(u_1)$. But suppose that there were elements of U not contained in $\psi_0(u_1)$. Then by 96 there would have to be among them a smallest number w ; and since, by what has just been said, this is different from the smallest number u_1 of the system U , by 117 there must also be in U a number v that is next smaller than w , from which it follows at the same time that $w = \psi(v)$. Since now $v < w$, by the definition of w the number v must certainly be contained in $\psi_0(u_1)$; from this, however, by 55 it follows that also $\psi(v)$, hence w , must be contained in $\psi_0(u_1)$, and since this contradicts the definition of w , our preceding supposition is inadmissible. Hence $U \subset \psi_0(u_1)$ and consequently also $U = \psi_0(u_1)$, as asserted. From $\alpha, \beta, \gamma, \delta$ it follows now by 71 that U is a simply infinite system ordered by ψ , as was to be proved.

123. Proposition. In consequence of 121 and 122, any part T of the sequence of numbers N is finite or simply infinite according as there is, or is not, a greatest number in T .

§ 9. Definition of a mapping of the sequence of numbers by induction

124. In what follows we shall also denote numbers by small Latin letters and shall keep, in general, all notation from the preceding §§ 6 to 8, while Ω denotes an arbitrary system whose elements need not be contained in N .

125. Proposition. If an arbitrary (similar or non-similar) mapping θ of a system Ω into itself is given, and in addition a definite element ω in Ω is given, then to every number n there corresponds one and only one mapping ψ_n of the corresponding number-system Z_n defined in 98, satisfying the conditions¹

- I. $\psi_n(Z_n) \subset \Omega$,
- II. $\psi_n(1) = \omega$,
- III. $\psi_n(t') = \theta\psi_n(t)$, if $t < n$,

where the sign $\theta\psi_n$ has the meaning given in 25.

Proof by complete induction (80). For

ρ . the proposition is true for $n = 1$. In this case, by 102, the system Z_n consists of the single number 1, and therefore the mapping ψ_1 is already completely, and in such a way, defined by II that I is fulfilled, while III is wholly absent.

σ . If the proposition is true for a number n , we show that it is also true for the following number $p = n'$, and we begin by proving that there can be only one corresponding mapping ψ_p of the system Z_p . Indeed, suppose that a mapping ψ_p satisfies the conditions

- I'. $\psi_p(Z_p) \subset \Omega$,
- II'. $\psi_p(1) = \omega$,
- III'. $\psi_p(m') = \theta\psi_p(m)$, if $m < p$.

Then, since $Z_n \subset Z_p$ by 107, this mapping contains, by 21, also a mapping of Z_n , which evidently satisfies the same conditions I, II, III as ψ_n and consequently agrees

¹For clarity I have here and in the following proposition 126 stated condition I separately, although it is really already a consequence of II and III.

entirely with ψ_n ; hence for all numbers m contained in Z_n , that is, by 98, for all numbers m that are $< p$, i.e. $\leq n$, one must have

$$\psi_p(m) = \psi_n(m) \quad (m)$$

and as a special case also

$$\psi_p(n) = \psi_n(n). \quad (n)$$

Since, furthermore, by 105 and 108, p is the only number of the system Z_p not contained in Z_n , and since by III' and (n) also

$$\psi_p(p) = \theta\psi_n(n) \quad (p)$$

must hold, the correctness of our assertion follows: there can be only one mapping ψ_p of the system Z_p satisfying I', II', III', because ψ_p has been completely reduced to ψ_n by the just derived conditions (m) and (p). We have now to show conversely that this mapping ψ_p of the system Z_p , completely determined by (m) and (p), really satisfies I', II', III'. Evidently I' follows from (m) and (p), with regard to I and to the fact that $\theta(\Omega) \subset \Omega$. Likewise II' follows from (m) and II, since by 99 the number 1 is contained in Z_n . The correctness of III' follows first for those numbers m that are $< n$ from (m) and III, and for the only remaining number $m = n$ it follows from (p) and (n). This proves completely that from the validity of our proposition for the number n there always follows its validity for the following number p , as was to be proved.

126. Proposition of definition by induction. If an arbitrary (similar or non-similar) mapping θ of a system Ω into itself is given, and in addition a definite element ω in Ω is given, then there is one and only one mapping ψ of the sequence of numbers N satisfying the conditions

- I. $\psi(N) \subset \Omega$,
- II. $\psi(1) = \omega$,
- III. $\psi(n') = \theta\psi(n)$,

where n denotes any number.

Proof. If such a mapping ψ really exists, then by 21 it contains a mapping ψ_n of the system Z_n satisfying the conditions I, II, III stated in 125. Since there is always one and only one such mapping ψ_n , necessarily

$$\psi(n) = \psi_n(n) \quad (n)$$

must hold. Since ψ is thereby completely determined, it follows that there can be only one such mapping ψ (cf. the end of 130). Conversely, that the mapping ψ determined by (n) also satisfies our conditions I, II, III follows easily from (n), taking account of the properties I, II, and (p) proved in 125, as was to be proved.

127. Proposition. Under the hypotheses made in the preceding proposition,

$$\psi(T') = \theta\psi(T),$$

where T denotes any part of the sequence of numbers N .

Proof. For if t denotes every number of the system T , then $\psi(T')$ consists of all elements $\psi(t')$, and $\theta\psi(T)$ consists of all elements $\theta\psi(t)$; our proposition follows from this, since by III in 126, $\psi(t') = \theta\psi(t)$.

128. Proposition. Keeping the same hypotheses, and denoting by θ_0 the chains (44) that correspond to the mapping θ of the system Ω into itself, one has

$$\psi(N) = \theta_0(\omega).$$

Proof. First we show by complete induction (80) that

$$\psi(N) \subset \theta_0(\omega),$$

that is, that every image $\psi(n)$ is also an element of $\theta_0(\omega)$. Indeed,

ρ . this proposition is true for $n = 1$, because by 126. II $\psi(1) = \omega$, and by 45 $\omega \subset \theta_0(\omega)$.

σ . If the proposition is true for a number n , that is, if $\psi(n) \subset \theta_0(\omega)$, then by 55 also $\theta(\psi(n)) \subset \theta_0(\omega)$, that is, by 126. III, $\psi(n') \subset \theta_0(\omega)$; hence the proposition is also true for the following number n' , as was to be proved.

In order next to prove that every element v of the chain $\theta_0(\omega)$ is contained in $\psi(N)$, that is, that

$$\theta_0(\omega) \subset \psi(N),$$

we again apply complete induction, namely proposition 59 transferred to Ω and the mapping θ . Indeed,

ρ . the element ω is $= \psi(1)$, and hence is contained in $\psi(N)$.

σ . If v is a common element of the chain $\theta_0(\omega)$ and of the system $\psi(N)$, then $v = \psi(n)$, where n denotes a number; from this it follows by 126. III that $\theta(v) = \theta\psi(n) = \psi(n')$, so that $\theta(v)$ is also contained in $\psi(N)$, as was to be proved.

From the proved propositions $\psi(N) \subset \theta_0(\omega)$ and $\theta_0(\omega) \subset \psi(N)$ it follows by 5 that $\psi(N) = \theta_0(\omega)$, as was to be proved.

129. Proposition. Under the same hypotheses, generally

$$\psi(n_0) = \theta_0(\psi(n)).$$

Proof by complete induction 80. For

ρ . the proposition holds by 128 for $n = 1$, since $1_0 = N$ and $\psi(1) = \omega$.

σ . If the proposition is true for a number n , then

$$\theta(\psi(n_0)) = \theta(\theta_0(\psi(n)));$$

now by 127 and 75

$$\theta(\psi(n_0)) = \psi(n'_0),$$

and by 57 and 126. III

$$\theta(\theta_0(\psi(n))) = \theta_0(\theta(\psi(n))) = \theta_0(\psi(n'));$$

hence

$$\psi(n'_0) = \theta_0(\psi(n')),$$

that is, the proposition is also true for the number n' following n , as was to be proved.

130. Remark. Before passing to the most important applications of proposition 126, proved above, on definition by induction (§§ 10 to 14), it is worth the trouble to call attention to a circumstance by which it differs essentially from the proposition of demonstration by induction proved in 80, or rather already in 59, 60, however close the kinship between the former and the latter may seem. Namely, while

proposition 59 holds quite generally for every chain A_0 , where A is any part of a system S mapped into itself by an arbitrary mapping φ (§ 4), the matter is quite different with proposition 126, which asserts only the existence of a contradiction-free, or single-valued, mapping ψ of the simply infinite system 1_0 . If in the latter proposition, while retaining the hypotheses on Ω and θ , one wished to put in place of the sequence of numbers 1_0 an arbitrary chain A_0 from such a system S , and to define a mapping ψ of A_0 into Ω in a manner similar to 126. II, III, by requiring that

ρ . to every element a of A there correspond a definite element $\psi(a)$ chosen from Ω , and

σ . for every element n contained in A_0 and its image $n' = \varphi(n)$, the condition $\psi(n') = \theta\psi(n)$ should hold,

then very often the case would occur that no such mapping ψ exists at all, because these conditions ρ, σ can come into contradiction with one another even if the freedom of choice contained in ρ is restricted in advance in accordance with condition σ . An example will suffice to become convinced of this. Let the system S consisting of the two different elements a and b be mapped by φ into itself in such a way that $a' = b, b' = a$; then evidently $a_0 = b_0 = S$. Further, let the system Ω consisting of the three different elements α, β, γ be mapped by θ into itself so that $\theta(\alpha) = \beta, \theta(\beta) = \gamma, \theta(\gamma) = \alpha$. If one now requires a mapping ψ of a_0 into Ω such that $\psi(a) = \alpha$ and in addition, for every element n contained in a_0 , always $\psi(n') = \theta\psi(n)$, then one meets a contradiction; for with $n = a$ one obtains $\psi(b) = \theta(\alpha) = \beta$, and from this, with $n = b$, it would follow that $\psi(a) = \theta(\beta) = \gamma$, whereas $\psi(a) = \alpha$ had been required.

But if there is a mapping ψ of A_0 into Ω satisfying the above conditions ρ, σ without contradiction, then it follows easily from 60 that it is completely determined. For if the mapping χ satisfies the same conditions, then generally $\chi(n) = \psi(n)$, because by ρ this proposition holds for all elements $n = a$ contained in A , and because, if it holds for an element n of A_0 , then by σ it must also hold for its image n' .

131. In order to bring to light the scope of our proposition 126, we insert here a consideration that is also useful for other investigations, for example for the so-called theory of groups.

We consider a system Ω whose elements admit a certain composition in such a way that from an element ν , by the action of an element ω , there always arises again a definite element of the same system Ω , which may be denoted by $\omega \cdot \nu$ or $\omega\nu$, and is in general to be distinguished from $\nu\omega$. One may also understand this by saying that to every definite element ω there corresponds a definite mapping of the system Ω into itself, perhaps denoted by ω , insofar as every element ν yields the definite image $\omega(\nu) = \omega\nu$. Applying proposition 126 to this system Ω and its element ω , while at the same time replacing the mapping there denoted by θ with ω , to every number n there corresponds a definite element $\psi(n)$ contained in Ω , which may now be denoted by the symbol ω^n and is sometimes called the n -th power of ω . This concept is completely explained by the conditions imposed on it,

$$\text{II. } \omega^1 = \omega, \quad \text{III. } \omega^{n'} = \omega\omega^n,$$

and its existence is secured by the proof of proposition 126.

If the preceding composition of elements is, in addition, such that for arbitrary elements μ, ν, ω one always has $\omega(\nu\mu) = (\omega\nu)\mu$, then also the propositions

$$\omega^{n'} = \omega^n \omega, \quad \omega^m \omega^n = \omega^n \omega^m$$

hold; their proofs may easily be carried out by complete induction (80), and may be left to the reader.

The preceding general consideration can be applied immediately to the following example. If S is a system of arbitrary elements, and Ω the corresponding system whose elements are all mappings ν of S into itself (36), then these elements can always be composed by 25, because $\nu(S) \subset S$, and the mapping $\omega\nu$ composed from such mappings ν and ω is itself again an element of Ω . Then all elements ω^n are also mappings of S into itself, and one says that they arise by repetition of the mapping ω . We shall now point out a simple connection between this concept and the concept of the chain $\omega_0(A)$ explained in 44, where A again denotes any part of S . For brevity, let the image $\omega^n(A)$ produced by the mapping ω^n be denoted by A_n ; then it follows from III and 25 that $\omega(A_n) = A_{n'}$. From this it follows easily by complete induction (80) that all these systems A_n are parts of the chain $\omega_0(A)$; for ρ . this assertion holds by 50 for $n = 1$, and

σ . if it holds for a number n , then from 55 and from $A_{n'} = \omega(A_n)$ it follows that it also holds for the following n' , as was to be proved. Since further, by 45, $A \subset \omega_0(A)$, it follows from 10 that also the system K composed of A and all the images A_n is part of $\omega_0(A)$. Conversely, since by 23 $\omega(K)$ is composed of $\omega(A) = A_1$ and all systems $\omega(A_n) = A_{n'}$, hence by 78 of all systems A_n , which by 9 are parts of K , it follows by 10 that $\omega(K) \subset K$, that is, K is a chain (37); and since by 9 $A \subset K$, by 47 also $\omega_0(A) \subset K$. Hence $\omega_0(A) = K$, that is, the following proposition holds: if ω is a mapping of a system S into itself and A is any part of S , then the chain of A corresponding to the mapping ω is composed of A and all images $\omega^n(A)$ arising by repetition of ω . We recommend that the reader return to the earlier propositions 57, 58 with this conception of a chain.

§ 10. The class of simply infinite systems

132. Proposition. All simply infinite systems are similar to the sequence of numbers N , and consequently by 33 also to one another.

Proof. Let the simply infinite system Ω be ordered by the mapping θ (71), and let ω be the fundamental element of Ω occurring here. If, again, we denote by θ_0 the chains corresponding to the mapping θ (44), then by 71 the following hold:

- α . $\theta(\Omega) \subset \Omega$,
- β . $\Omega = \theta_0(\omega)$,
- γ . ω is not contained in $\theta(\Omega)$,
- δ . the mapping θ is similar.

Let ψ denote the mapping of the sequence of numbers N defined in 126. Then from β and 128 first follows

$$\psi(N) = \Omega,$$

and therefore by 32 we have only to show that ψ is a similar mapping, that is, by 26, that to different numbers m, n there correspond different images $\psi(m), \psi(n)$. For symmetry's sake we may by 90 assume that $m > n$, hence $m \subset n'_0$, and the proposition to be proved amounts to saying that $\psi(n)$ is not contained in $\psi(n'_0)$, and therefore by 127 not contained in $\theta\psi(n_0)$. We prove this for every number n by complete induction (80). Indeed,

ρ . this proposition holds by γ for $n = 1$, because $\psi(1) = \omega$ and $\psi(1_0) = \psi(N) = \Omega$.

σ . If the proposition is true for a number n , then it is also true for the following number n' ; for if $\psi(n')$, that is, $\theta\psi(n)$, were contained in $\theta\psi(n'_0)$, then by δ and 27, $\psi(n)$ would also have to be contained in $\psi(n'_0)$, whereas our assumption says exactly the contrary, as was to be proved.

133. Proposition. Every system that is similar to a simply infinite system, and consequently by 132 and 33 also to the sequence of numbers N , is simply infinite.

Proof. If Ω is a system similar to the sequence of numbers N , then by 32 there is a similar mapping ψ of N such that

$$\text{I. } \psi(N) = \Omega$$

and we set

$$\text{II. } \psi(1) = \omega.$$

If by 26 $\bar{\psi}$ denotes the inverse, likewise similar, mapping of Ω , then to every element ν of Ω there corresponds a definite number $\bar{\psi}(\nu) = n$, namely the one whose image is $\psi(n) = \nu$. Since to this number n there corresponds a definite following number $\varphi(n) = n'$, and to this again a definite element $\psi(n')$ in Ω , to every element ν of the system Ω there belongs a definite element $\psi(n')$ of the same system, which we shall denote as the image of ν by $\theta(\nu)$. This completely determines a mapping θ of Ω into itself²; and to prove our proposition we shall show that Ω is ordered by θ as a simply infinite system (71), that is, that all the conditions $\alpha, \beta, \gamma, \delta$ stated in the proof of 132 are fulfilled. First α is immediately clear from the definition of θ . Since further to every number n there corresponds an element $\nu = \psi(n)$ for which $\theta(\nu) = \psi(n')$, one has generally

$$\text{III. } \psi(n') = \theta\psi(n),$$

and from this, in connection with I, II, α , it follows that the mappings θ, ψ satisfy all conditions of proposition 126; hence β follows from 128 and I. By 127 and I, furthermore,

$$\psi(N') = \theta\psi(N) = \theta(\Omega),$$

and from this, in connection with II and the similarity of ψ , γ follows, because otherwise $\psi(1)$ would be contained in $\psi(N')$, and hence by 27 the number 1 would have to be contained in N' , which by 71. γ is not the case. Finally, if μ, ν are elements of Ω , and m, n denote the corresponding numbers whose images are $\psi(m) = \mu$, $\psi(n) = \nu$, then from the assumption $\theta(\mu) = \theta(\nu)$ it follows, by what has just been said, that $\psi(m') = \psi(n')$; from this, because ψ, φ are similar, it follows that $m' = n'$, $m = n$, and therefore also $\mu = \nu$. Hence δ also holds, as was to be proved.

134. Remark. In consequence of the two preceding propositions 132, 133, all simply infinite systems form a class in the sense of 34. At the same time, with regard to 71 and 73, it is clear that every proposition about the numbers, that is, about the elements n of the simply infinite system N ordered by the mapping φ , and indeed every such proposition in which the special nature of the elements n is wholly disregarded and only concepts arising from the ordering φ are involved, possesses completely general validity also for every other simply infinite system Ω ordered by a mapping θ and for its elements ν . The transfer from N to Ω (for example, also the translation of an arithmetical proposition from one language into another) is effected by the mapping ψ considered in 132, 133, which transforms every element n of N into an element ν of Ω , namely into $\psi(n)$. This element ν may be called the n -th element of Ω , and accordingly the number n itself is the n -th

²Evidently θ is the mapping $\psi\varphi\bar{\psi}$ composed from $\bar{\psi}$, φ , ψ according to 25.

number of the sequence of numbers N . The same significance that the mapping φ has for the laws in the domain N , insofar as to every element n there follows a definite element $\varphi(n) = n'$, belongs, after the transformation effected by ψ , to the mapping θ for the same laws in the domain Ω , insofar as to the element $\nu = \psi(n)$ produced by transformation of n there follows the element $\theta(\nu) = \psi(n')$ produced by transformation of n' . Thus one may rightly say that φ is transformed by ψ into θ , which is expressed symbolically by $\theta = \psi\varphi\bar{\psi}$, $\varphi = \bar{\psi}\theta\psi$. By these remarks, I believe, the explanation of the concept of numbers given in 73 is completely justified. We now pass to further applications of proposition 126.

§ 11. Addition of numbers

135. Definition. It is natural to apply the definition, presented in proposition 126, of a mapping ψ of the sequence of numbers N , or of the function $\psi(n)$ thereby determined, to the case in which the system there denoted by Ω , in which the image $\psi(N)$ is to be contained, is the sequence of numbers N itself; for this system Ω already has a mapping θ of Ω into itself, namely the mapping φ by which N is ordered as a simply infinite system (71, 73). Then $\Omega = N$, $\theta(n) = \varphi(n) = n'$, and hence

$$\text{I. } \psi(N) \subset N.$$

It remains, in order to determine ψ completely, only to choose the element ω from Ω , that is, from N , at will. If we take $\omega = 1$, then ψ is evidently the identical mapping (21) of N , because the conditions

$$\psi(1) = 1, \quad \psi(n') = (\psi(n))'$$

are generally satisfied by $\psi(n) = n$. If, therefore, another mapping ψ of N is to be generated, then for ω one must choose a number m' different from 1 and contained by 78 in N' , where m itself denotes an arbitrary number. Since the mapping ψ evidently depends on the choice of this number m , we denote the corresponding image $\psi(n)$ of an arbitrary number n by the symbol $m + n$ and call this number the sum that arises from the number m by addition of the number n , or briefly the sum of the numbers m, n . It is therefore completely determined by 126 by the conditions³

$$\text{II. } m + 1 = m', \quad \text{III. } m + n' = (m + n)'$$

136. Proposition. One has $m' + n = m + n'$.

Proof by complete induction (80). For

ρ . the proposition is true for $n = 1$, because by 135. II

$$m' + 1 = (m')' = (m + 1)'$$

and by 135. III, $(m + 1)' = m + 1'$.

σ . If the proposition is true for a number n , and if one sets the following number $n' = p$, then $m' + n = m + p$, hence also $(m' + n)' = (m + p)'$; from this, by 135. III,

³The preceding definition of addition, founded directly on proposition 126, seems to me the simplest. With the aid of the concept developed in 131, however, one can also define the sum $m + n$ by $\varphi^n(m)$ or also by $\varphi^m(n)$, where φ again has the above meaning. To prove the complete agreement of these definitions with the preceding one, it is only necessary, by 126, to show that, if $\varphi^n(m)$ or $\varphi^m(n)$ is denoted by $\psi(n)$, the conditions $\psi(1) = m'$, $\psi(n') = \varphi\psi(n)$ are fulfilled; this is easily done with the aid of complete induction (80), using 131.

$m' + p = m + p'$ follows. Thus the proposition holds also for the following number p , as was to be proved.

137. Proposition. One has $m' + n = (m + n)'$.

The proof follows from 136 and 135. III.

138. Proposition. One has $1 + n = n'$.

Proof by complete induction (80). For

ρ . the proposition is true by 135. II for $n = 1$.

σ . If the proposition holds for a number n , and if $n' = p$, then $1 + n = p$, hence also $(1 + n)' = p'$; therefore by 135. III, $1 + p = p'$, that is, the proposition holds also for the following number p , as was to be proved.

139. Proposition. One has $1 + n = n + 1$.

The proof follows from 138 and 135. II.

140. Proposition. One has $m + n = n + m$.

Proof by complete induction (80). For

ρ . the proposition is true by 139 for $n = 1$.

σ . If the proposition holds for a number n , then from it also follows $(m + n)' = (n + m)'$, that is, by 135. III, $m + n' = n + m'$; hence by 136, $m + n' = n' + m$. Thus the proposition holds also for the following number n' , as was to be proved.

141. Proposition. One has $(l + m) + n = l + (m + n)$.

Proof by complete induction (80). For

ρ . the proposition is true for $n = 1$, because by 135. II, III, II, $(l + m) + 1 = (l + m)' = l + m' = l + (m + 1)$.

σ . If the proposition holds for a number n , then from it also follows $((l + m) + n)' = (l + (m + n))'$, that is, by 135. III,

$$(l + m) + n' = l + (m + n)' = l + (m + n'),$$

so the proposition holds also for the following number n' , as was to be proved.

142. Proposition. One has $m + n > m$.

Proof by complete induction (80). For

ρ . the proposition is true by 135. II and 91 for $n = 1$.

σ . If the proposition holds for a number n , then by 95 it also holds for the following number n' , because by 135. III and 91

$$m + n' = (m + n)' > m + n,$$

as was to be proved.

143. Proposition. The conditions $m > a$ and $m + n > a + n$ are equivalent.

Proof by complete induction (80). For

ρ . the proposition holds by 135. II and 94 for $n = 1$.

σ . If the proposition holds for a number n , then it also holds for the following number n' , because by 94 the condition $m + n > a + n$ is equivalent to $(m + n)' > (a + n)'$, and therefore by 135. III also to

$$m + n' > a + n',$$

as was to be proved.

144. Proposition. If $m > a$ and $n > b$, then also

$$m + n > a + b.$$

Proof. From our hypotheses it follows by 143 that $m + n > a + n$ and $n + a > b + a$, or, what is the same by 140, $a + n > a + b$; from this the proposition follows by 95.

145. Proposition. If $m + n = a + n$, then $m = a$.

Proof. For if m is not $= a$, then by 90 either $m > a$ or $m < a$; correspondingly, by 143, $m + n > a + n$ or $m + n < a + n$. Hence by 90, $m + n$ certainly cannot be $= a + n$, as was to be proved.

146. Proposition. If $l > n$, then there is one and, by 145, only one number m satisfying the condition $m + n = l$.

Proof by complete induction (80). For

ρ . the proposition is true for $n = 1$. Indeed, if $l > 1$, that is, by 89, if l is contained in N' and hence is the image m' of a number m , then by 135. II it follows that $l = m + 1$, as was to be proved.

σ . If the proposition holds for a number n , we show that it also holds for the following number n' . Indeed, if $l > n'$, then by 91 and 95 also $l > n$, and consequently there is a number k satisfying $l = k + n$. Since by 138 this number is different from 1 (for otherwise $l = n'$), it is by 78 the image m' of a number m , and consequently $l = m' + n$, hence by 136 also $l = m + n'$, as was to be proved.

§ 12. Multiplication of numbers

147. Definition. After in the preceding § 11 an infinite system of new mappings of the sequence of numbers N into itself has been found, each of these may again be used, by 126, to generate further new mappings ψ of N . If there we put $\Omega = N$ and $\theta(n) = m + n = n + m$, where m is a definite number, then again certainly

$$\text{I. } \psi(N) \subset N,$$

and in order to determine ψ completely it remains only to choose the element ω from N at will. The simplest case occurs when this choice is brought into a certain agreement with the choice of θ , by putting $\omega = m$. Since the mapping ψ thereby completely determined depends on this number m , we denote the corresponding image $\psi(n)$ of an arbitrary number n by the symbol $m \times n$, or $m \cdot n$, or mn , and call this number the product that arises from the number m by multiplication by the number n , or briefly the product of the numbers m, n . It is therefore completely determined by 126 by the conditions

$$\text{II. } m \cdot 1 = m, \quad \text{III. } mn' = mn + m.$$

148. Proposition. One has $m'n = mn + n$.

Proof by complete induction (80). For

ρ . the proposition is true by 147. II and 135. II for $n = 1$.

σ . If the proposition holds for a number n , then

$$m'n + m' = (mn + n) + m',$$

and from this, by 147. III, 141, 140, 136, 141, 147. III,

$$\begin{aligned} m'n' &= mn + (n + m') = mn + (m' + n) \\ &= mn + (m + n') = (mn + m) + n' = mn' + n'; \end{aligned}$$

hence the proposition holds also for the following number n' , as was to be proved.

149. Proposition. One has $1 \cdot n = n$.

Proof by complete induction (80). For

ρ . the proposition is true by 147. II for $n = 1$.

σ . If the proposition holds for a number n , then $1 \cdot n + 1 = n + 1$, that is, by 147. III and 135. II, $1 \cdot n' = n'$; hence the proposition also holds for the following number n' , as was to be proved.

150. Proposition. One has $mn = nm$.

Proof by complete induction (80). For

ρ . the proposition holds by 147. II and 149 for $n = 1$.

σ . If the proposition holds for a number n , then

$$mn + m = nm + m,$$

that is, by 147. III and 148, $mn' = n'm$; hence the proposition holds also for the following number n' , as was to be proved.

151. Proposition. One has $l(m + n) = lm + ln$.

Proof by complete induction (80). For

ρ . the proposition is true by 135. II, 147. III, 147. II for $n = 1$.

σ . If the proposition holds for a number n , then

$$l(m + n) + l = (lm + ln) + l;$$

by 147. III and 135. III, however,

$$l(m + n) + l = l(m + n)' = l(m + n'),$$

and by 141 and 147. III,

$$(lm + ln) + l = lm + (ln + l) = lm + ln'.$$

Hence $l(m + n') = lm + ln'$, that is, the proposition holds also for the following number n' , as was to be proved.

152. Proposition. One has $(m + n)l = ml + nl$.

The proof follows from 151 and 150.

153. Proposition. One has $(lm)n = l(mn)$.

Proof by complete induction (80). For

ρ . the proposition holds by 147. II for $n = 1$.

σ . If the proposition holds for a number n , then

$$(lm)n + lm = l(mn) + lm,$$

that is, by 147. III, 151, 147. III,

$$(lm)n' = l(mn + m) = l(mn'),$$

so the proposition also holds for the following number n' , as was to be proved.

154. Remark. If in 147 no relation had been assumed between ω and θ , but instead $\omega = k$, $\theta(n) = m + n$ had been set, then by 126 a less simple mapping ψ of the sequence of numbers N would have arisen; for the number 1, $\psi(1) = k$, and for every other number, hence every number contained in the form n' , one would have $\psi(n') = mn + k$. For, as one easily convinces oneself with the aid of the preceding propositions, this fulfills the condition $\psi(n') = \theta\psi(n)$, that is, $\psi(n') = m + \psi(n)$, for all numbers n .

§ 13. Exponentiation of numbers

155. Definition. If in proposition 126 we again put $\Omega = N$, and further put $\omega = a$, $\theta(n) = an = na$, then there arises a mapping ψ of N which again satisfies the condition

$$\text{I. } \psi(N) \subset N.$$

We denote the corresponding image $\psi(n)$ of an arbitrary number n by the symbol a^n , and call this number a power of the base a , while n is called the exponent of this power of a . This concept is therefore completely determined by the conditions

$$\text{II. } a^1 = a, \quad \text{III. } a^{n'} = a \cdot a^n = a^n \cdot a.$$

156. Proposition. One has $a^{m+n} = a^m \cdot a^n$.

Proof by complete induction (80). For

ρ . the proposition holds by 135. II, 155. III, and 155. II for $n = 1$.

σ . If the proposition holds for a number n , then

$$a^{m+n} \cdot a = (a^m \cdot a^n)a;$$

but by 155. III and 135. III,

$$a^{m+n} \cdot a = a^{(m+n)'} = a^{m+n'},$$

and by 153 and 155. III,

$$(a^m \cdot a^n)a = a^m(a^n \cdot a) = a^m \cdot a^{n'}.$$

Thus $a^{m+n'} = a^m \cdot a^{n'}$, that is, the proposition also holds for the following number n' , as was to be proved.

157. Proposition. One has $(a^m)^n = a^{mn}$.

Proof by complete induction (80). For

ρ . the proposition holds by 155. II and 147. II for $n = 1$.

σ . If the proposition holds for a number n , then

$$(a^m)^n \cdot a^m = a^{mn} \cdot a^m;$$

but by 155. III,

$$(a^m)^n \cdot a^m = (a^m)^{n'},$$

and by 156 and 147. III,

$$a^{mn} \cdot a^m = a^{mn+m} = a^{mn'}.$$

Thus $(a^m)^{n'} = a^{mn'}$, that is, the proposition also holds for the following number n' , as was to be proved.

158. Proposition. One has $(ab)^n = a^n \cdot b^n$.

Proof by complete induction (80). For

ρ . the proposition holds by 155. II for $n = 1$.

σ . If the proposition holds for a number n , then by 150, 153, and 155. III also

$$(ab)^n \cdot a = a(a^n \cdot b^n) = (a \cdot a^n)b^n = a^{n'} \cdot b^n,$$

and from this

$$((ab)^n \cdot a)b = (a^{n'} \cdot b^n)b;$$

but by 153 and 155. III,

$$((ab)^n \cdot a)b = (ab)^n \cdot (ab) = (ab)^{n'},$$

and likewise

$$(a^{n'} \cdot b^n)b = a^{n'} \cdot (b^n \cdot b) = a^{n'} \cdot b^{n'}.$$

Thus $(ab)^{n'} = a^{n'} \cdot b^{n'}$, that is, the proposition also holds for the following number n' , as was to be proved.

§ 14. Number of elements of a finite system

159. Proposition. If Σ is an infinite system, then every one of the number systems Z_n explained in 98 can be similarly mapped into Σ (that is, is similar to a part of Σ), and conversely.

Proof. If Σ is infinite, then by 72 there is certainly a part T of Σ which is simply infinite, and hence by 132 is similar to the sequence of numbers N ; consequently, by 35, every system Z_n , as a part of N , is similar also to a part of T , and hence also to a part of Σ , as was to be proved.

The proof of the converse—however evident it may seem—is more complicated. If every system Z_n can be similarly mapped into Σ , then to every number n there corresponds such a similar mapping α_n of Z_n that $\alpha_n(Z_n) \subset \Sigma$. From the existence of such a series of mappings α_n , to be regarded as given but about which nothing further is assumed, we shall first derive, with the aid of proposition 126, the existence of a new series of such mappings ψ_n which has the special property that whenever $m \leq n$, and hence by 100 $Z_m \subset Z_n$, the mapping ψ_m of the part Z_m is contained in the mapping ψ_n of Z_n (21); that is, the mappings ψ_m and ψ_n agree completely with each other for all numbers contained in Z_m , and hence always also

$$\psi_m(m) = \psi_n(m).$$

In order to apply the cited proposition in accordance with this aim, let Ω mean the system whose elements are all possible similar mappings of all systems Z_n into Σ , and, with the aid of the given elements α_n , which are likewise contained in Ω , define in the following way a mapping θ of Ω into itself. If β is any element of Ω , hence, for example, a similar mapping of the definite system Z_n into Σ , then the system $\alpha_{n'}(Z_{n'})$ cannot be part of $\beta(Z_n)$; otherwise $Z_{n'}$ would, by 35, be similar to a part of Z_n , and hence by 107 to a proper part of itself, and consequently would be infinite, contrary to proposition 119. There are therefore certainly in $Z_{n'}$ one or several numbers p such that $\alpha_{n'}(p)$ is not contained in $\beta(Z_n)$. Of these numbers p we always choose—only in order to fix something definite—the least k (96), and, since by 108 $Z_{n'}$ is composed of Z_n and n' , we define a mapping γ of $Z_{n'}$ by requiring that for all numbers m contained in Z_n the image $\gamma(m) = \beta(m)$, and in addition that $\gamma(n') = \alpha_{n'}(k)$; this evidently similar mapping γ of $Z_{n'}$ into Σ we now regard as an image $\theta(\beta)$ of the mapping β , and in this way a mapping θ of the system Ω into itself is completely defined. After the things denoted in 126 by Ω and θ have been determined, we finally choose, for the element of Ω denoted there by ω , the given mapping α_1 ; by 126 this determines a mapping ψ of the sequence of numbers N into Ω , which, if we denote the corresponding image of an arbitrary number n not by $\psi(n)$ but by ψ_n , satisfies the conditions

$$\text{II. } \psi_1 = \alpha_1, \quad \text{III. } \psi_{n'} = \theta(\psi_n).$$

By complete induction (80) it first follows that ψ_n is a similar mapping of Z_n into Σ ; for

ρ . this is true by II for $n = 1$, and

σ . if this assertion is true for a number n , then from III and from the nature of the transition θ from β to γ described above it follows that the assertion is also true for the following number n' , as was to be proved.

Next we prove, again by complete induction (80), that, if m is any number, the special property announced above,

$$\psi_n(m) = \psi_m(m),$$

really belongs to all numbers n which are $\geq m$, and hence by 93 and 74 belong to the chain m_0 ; indeed,

ρ . this is immediately evident for $n = m$, and

σ . if this property belongs to a number n , then again from III and from the nature of θ it follows that it also belongs to the number n' , as was to be proved.

After this special property of our new series of mappings ψ_n has also been established, we can easily prove our proposition. We define a mapping χ of the sequence of numbers N by making the image $\chi(n) = \psi_n(n)$ correspond to each number n ; evidently (by 21) all mappings ψ_n are contained in this single mapping χ . Since ψ_n was a mapping of Z_n into Σ , it follows first that the sequence of numbers N is likewise mapped by χ into Σ , hence $\chi(N) \subset \Sigma$. If, further, m, n are different numbers, then by symmetry we may by 90 assume that $m < n$; then by the above $\chi(m) = \psi_m(m) = \psi_n(m)$ and $\chi(n) = \psi_n(n)$. But since ψ_n was a similar mapping of Z_n into Σ , and m, n are different elements of Z_n , $\psi_n(m)$ is different from $\psi_n(n)$, and hence also $\chi(m)$ is different from $\chi(n)$; that is, χ is a similar mapping of N . Since moreover N is an infinite system (71), the same holds by 67 for the system $\chi(N)$ similar to it, and by 68, because $\chi(N)$ is part of Σ , also for Σ , as was to be proved.

160. Proposition. A system Σ is finite or infinite according as there exists or does not exist a system Z_n similar to it.

Proof. If Σ is finite, then by 159 there are systems Z_n which cannot be similarly mapped into Σ ; since by 102 the system Z_1 consists of the single number 1 and consequently can be similarly mapped into every system, the least number k (96) to which there corresponds a system Z_k not similarly mappable into Σ must be different from 1, hence by 78 must be $= n'$, and since $n < n'$ (91), there is a similar mapping ψ of Z_n into Σ . If now $\psi(Z_n)$ were only a proper part of Σ , so that there were an element α in Σ which is not contained in $\psi(Z_n)$, then, since $Z_{n'} = \mathfrak{M}(Z_n, n')$ (108), one could extend this mapping ψ to a similar mapping ψ of $Z_{n'}$ into Σ by setting $\psi(n') = \alpha$, whereas by our assumption $Z_{n'}$ cannot be similarly mapped into Σ . Therefore $\psi(Z_n) = \Sigma$, that is, Z_n and Σ are similar systems. Conversely, if a system Σ is similar to a system Z_n , then by 119 and 67 Σ is finite, as was to be proved.

161. Definition. If Σ is a finite system, then by 160 there is one, and by 120 and 33 only one, number n to which there corresponds a system Z_n similar to the system Σ ; this number n is called the *number* of the elements contained in Σ (or also the *degree* of the system Σ), and one says that Σ consists of, or is, a system of n elements, or that the number n states how many elements are contained in Σ ⁴. When the numbers are used to express exactly this definite property of finite

⁴For clarity and simplicity we restrict, in what follows, the concept of number entirely to finite systems; therefore, when we speak of a number of certain things, it should always already be expressed thereby that the system whose elements these things are is finite.

systems, they are called *cardinal numbers*. As soon as a definite similar mapping ψ of the system Z_n is chosen, by virtue of which $\psi(Z_n) = \Sigma$, to every number m contained in Z_n (that is, to every number m which is $\leq n$) there corresponds a definite element $\psi(m)$ of the system Σ , and conversely, by 26, to every element of Σ there corresponds through the inverse mapping $\bar{\psi}$ a definite number m in Z_n . Very often all elements of Σ are denoted by a single letter, for example α , to which the distinguishing numbers m are attached as subscripts, so that $\psi(m)$ is denoted by α_m . One also says that these elements have been counted and ordered in a definite way by ψ , and calls α_m the m -th element of Σ ; if $m < n$, then $\alpha_{m'}$ is called the element following α_m , and α_n is called the last element. In this counting of the elements, therefore, the numbers m again occur as ordinal numbers (73).

162. Proposition. All systems similar to a finite system possess the same number of elements.

The proof follows immediately from 33 and 161.

163. Proposition. The number of the elements contained in Z_n , that is, of those numbers which are $\leq n$, is n .

Proof. For by 32 Z_n is similar to itself.

164. Proposition. If a system consists of a single element, then the number of its elements is $= 1$, and conversely.

The proof follows immediately from 2, 26, 32, 102, and 161.

165. Proposition. If T is a proper part of a finite system Σ , then the number of the elements of T is smaller than that of the elements of Σ .

Proof. By 68, T is a finite system, hence similar to a system Z_m , where m denotes the number of the elements of T ; further, if n is the number of the elements of Σ , so that Σ is similar to Z_n , then by 35 T is similar to a proper part E of Z_n , and by 33 also Z_m and E are similar to one another. If now $n \leq m$, hence $Z_n \subset Z_m$, then by 7 E would also be a proper part of Z_m , and consequently Z_m would be an infinite system, which contradicts proposition 119; hence (by 90) $m < n$, as was to be proved.

166. Proposition. If $\Gamma = \mathfrak{M}(B, \gamma)$, where B is a system of n elements and γ denotes an element of Γ not contained in B , then Γ consists of n' elements.

Proof. For if $B = \psi(Z_n)$, where ψ denotes a similar mapping of Z_n , then by 105 and 108 this mapping can be extended to a similar mapping ψ of $Z_{n'}$ by setting $\psi(n') = \gamma$, and in fact $\psi(Z_{n'}) = \Gamma$, as was to be proved.

167. Proposition. If γ is an element of a system Γ consisting of n' elements, then n is the number of all the other elements of Γ .

Proof. For if B denotes the collection of all elements in Γ different from γ , then $\Gamma = \mathfrak{M}(B, \gamma)$; if now b is the number of the elements of the finite system B , then by the preceding proposition b' is the number of the elements of Γ , hence $= n'$, from which by 26 it follows also that $b = n$, as was to be proved.

168. Proposition. If A consists of m and B consists of n elements, and if A and B have no common element, then $\mathfrak{M}(A, B)$ consists of $m + n$ elements.

Proof by complete induction (80). For

ρ . the proposition is true for $n = 1$ by 166, 164, and 135. II.

σ . If the proposition holds for a number n , then it holds also for the following number n' . In fact, if Γ is a system of n' elements, then by 167 one may put $\Gamma = \mathfrak{M}(B, \gamma)$, where γ is an element and B denotes the system of the n other elements of Γ . Now if A is a system of m elements, each of which is not contained in Γ , and hence is not contained in B either, and if one puts $\mathfrak{M}(A, B) = \Sigma$, then by our assumption $m + n$ is the number of the elements of Σ ; and since γ is not contained in Σ , by

166 the number of the elements contained in $\mathfrak{M}(\Sigma, \gamma)$ is $= (m + n)'$, hence (by 135. III) $= m + n'$. But since by 15 evidently $\mathfrak{M}(\Sigma, \gamma) = \mathfrak{M}(A, B, \gamma) = \mathfrak{M}(A, \Gamma)$, $m + n'$ is the number of the elements of $\mathfrak{M}(A, \Gamma)$, as was to be proved.

169. Proposition. If A, B are finite systems of respectively m, n elements, then $\mathfrak{M}(A, B)$ is a finite system, and the number of its elements is $\leq m + n$.

Proof. If $B \subset A$, then $\mathfrak{M}(A, B) = A$, and the number m of the elements of this system is (by 142) $< m + n$, as was asserted. If, however, B is not a part of A , and T is the system of all those elements of B which are not contained in A , then by 165 their number $p \leq n$, and since evidently

$$\mathfrak{M}(A, B) = \mathfrak{M}(A, T),$$

by 143 the number $m + p$ of the elements of this system is $\leq m + n$, as was to be proved.

170. Proposition. Every system composed of a number n of finite systems is finite.

Proof by complete induction (80). For

ρ . the proposition is self-evident by 8 for $n = 1$.

σ . If the proposition holds for a number n , and if Σ is composed of n' finite systems, let A be one of these systems and B the system composed of all the remaining ones; since their number is (by 167) $= n$, by our assumption B is a finite system. Since now evidently $\Sigma = \mathfrak{M}(A, B)$, it follows from this and from 169 that Σ is also a finite system, as was to be proved.

171. Proposition. If ψ is a non-similar mapping of a finite system Σ of n elements, then the number of the elements of the image $\psi(\Sigma)$ is smaller than n .

Proof. From all those elements of Σ which have one and the same image, choose only a single one each time, arbitrarily; then the system T of all these selected elements is evidently a proper part of Σ , because ψ is a non-similar mapping of Σ (26). At the same time, however, it is clear that the mapping of this part T contained in ψ (by 21) is similar, and that $\psi(T) = \psi(\Sigma)$. Hence the system $\psi(\Sigma)$ is similar to the proper part T of Σ , and from this our proposition follows by 162 and 165.

172. Final remark. Although it has just been proved that the number m of the elements of $\psi(\Sigma)$ is smaller than the number n of the elements of Σ , in many cases one nevertheless likes to say that the number of the elements of $\psi(\Sigma)$ is $= n$. Naturally the word number is then used in a sense different from the one used so far (161); namely, if α is an element of Σ , and a is the number of all those elements of Σ which have one and the same image $\psi(\alpha)$, then the latter, as an element of $\psi(\Sigma)$, is often still regarded as representative of a elements, which at least according to their origin may be regarded as different from one another, and accordingly is counted as an a -fold element of $\psi(\Sigma)$. In this way one arrives at the concept, very useful in many cases, of systems in which every element is furnished with a certain multiplicity number, which indicates how often it is to be counted as an element of the system. In the above case, for example, one would say that n is the number of the elements of $\psi(\Sigma)$ counted in this sense, while the number m of the really different elements of this system agrees with the number of the elements of T . Similar deviations from the original meaning of a technical term, which are nothing other than extensions of the original concepts, occur very frequently in mathematics; but it is not the purpose of this writing to go into this more closely.

Explanatory note to the preceding paper*

“What are and what should the numbers be?” became pathbreaking in two directions: for foundational research and for axiomatic set theory. Hilbert has only recently again pointed to its significance for foundational research (Math. Ann. 104); a detailed analysis of the writing by E. Zermelo is found in Landau’s obituary notice (Gött. Nachr. 1917). How strongly axiomatic set theory is influenced by Dedekind is shown by a comparison with the Zermelo axioms (Math. Ann. 65), which in part are taken directly from Dedekind’s “definitions” (§ 1 of the writing). It is known that in this connection the “axiom of infinity” had to be postulated, since Dedekind’s attempted proof (66) rests on the contradictory concept of the “set of everything thinkable”; likewise, it is known that the axiom of choice enters into Dedekind’s considerations (159). Zermelo’s second proof of the well-ordering theorem can also be regarded as a transfer of the proof given here for the possibility of complete induction to transfinite induction; in the transfinite case, however, the axiom of choice already had to be added here to the other axioms implicitly used by Dedekind. Dedekind was able to avoid it for ordinary complete induction because he had at his disposal the mapping entering into the definition of the infinite. The theorem of “definition” by complete induction (126), going beyond the “proof” by complete induction, is sharply worked out for the transfinite by J. v. Neumann (Math. Ann. 99). The theorem is used in particular in the algebra of infinite domains, just as Dedekind obtains the rules of calculation for the whole numbers by means of definition by complete induction.

Noether.

*Cf. also the explanatory notes to LX and LXII.